

How do trees grow in Western North America: A new modeling approach

1 The motivations behind the project

💡 The technical motivation

Bayesian magic! We fit all together the low frequency, decadal (to centennial) variations (the *detrending*) **and** the climatic variations (of interest). We let the model and the data speak, i.e. disentangle allometry (age/size trends in growth) from climate. This way, we avoid many arbitrary modeling decision (e.g. when one has to choose the frequency cut-off of a spline). And all the uncertainties get propagated to any decisions/predictions.

💡 The ecological motivation

Tree ring data offers offer an unique opportunity to learn how trees grow in a changing climate, through the complex interactions between external events and size-dependent developmental stages. While dendrochronology has mostly focus on past climate reconstructions, our approach do not try to standardize away the complexities of individual-level growth and species-specific variations. On the contrary, we believe that this complexity—if handle correctly by a model—can help us understand how tree growth varies within and **across species, ecosystems, and climates**. And *in fine* our approach could improve our forecasts of forest dynamics, carbon sequestration, and thus climate change itself!

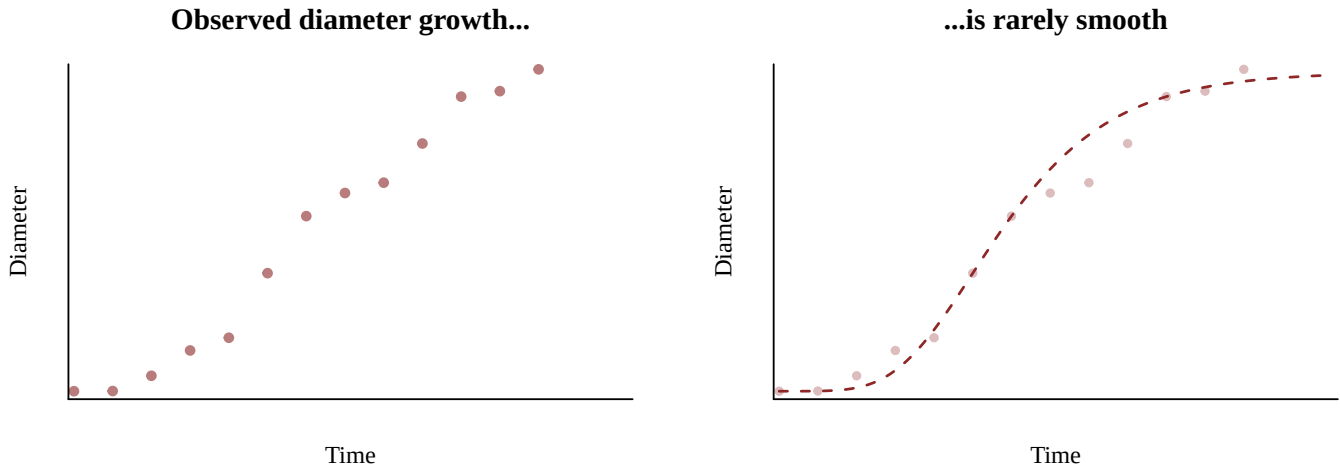
2 A quick summary of the model

Before diving into the results of our first inferences, let's describe the model!
We can divide the model in two main parts:

- the mid- to long-term processes that are not directly related to climate, such as **allometric constraints**, neighbour competition dynamics, ...
- the short-term variations that are related to external conditions, mostly **climate**, but also pest outbreaks, ...

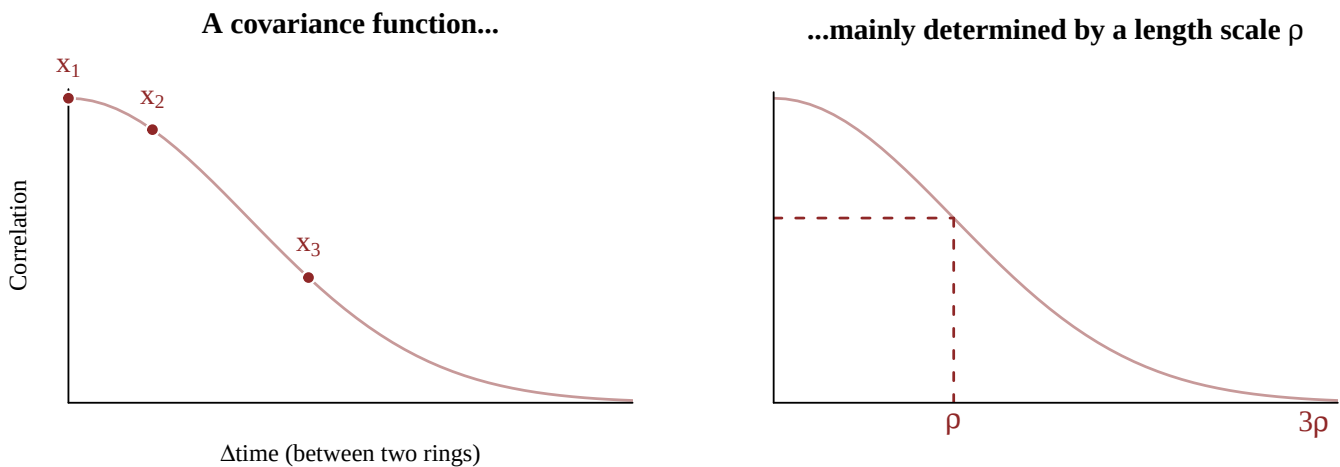
2.1 Mid- to long-term processes

When we think about these low frequency variations, we mainly think about the age/size trends that affect tree growth, due to the natural size and shape of trees. We want this part of the model to be **flexible** enough to accommodate phase shifts, since tree growth is rarely perfectly smooth.



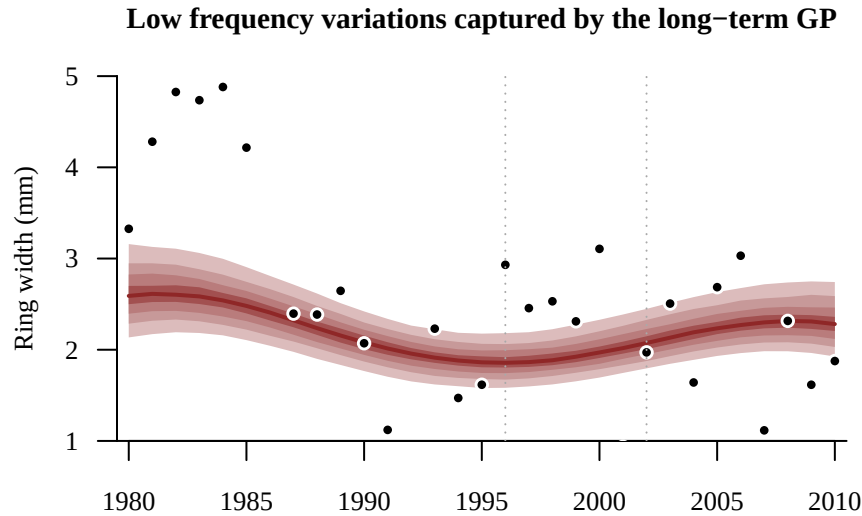
We believe that these variations are “*not directly related to climate*”, but not totally independent of the climate neither: a big storm could knock down surrounding trees and reduce competition for an individual.

We model these variations with a first **Gaussian process** (GP). The GP is characterized by a covariance function that captures the temporal correlation between ring widths that persists over one or several decades. The **length scale** of this covariance function determined how quickly the correlations between ring widths decay. In the example below, ring width x_2 is more related to x_1 than x_3 is.



The correlation across ring widths becomes negligible around 3ρ years.

This mid- to long-term GP allows to capture, at the individual tree level, the low frequency variations that appear in the ring width measurements.



Since we expect some difference across fast- or slow-growing species, we allow the length scale parameter to **vary across species**.

2.2 Short-term variations

When we think about these low frequency variations, we mainly think about the yearly climatic conditions that explain short-term variations in growth. We first believed that these climatic effects were shared across conspecific trees within the same stand—in other words, that the trees from the same species on the same stand would have the same decrease/increase in growth because they experience similar climatic conditions. But we will see that it's a bit more complex than!

2.2.1 Climatic predictors (stand-level)

This part of the model is quite simple.

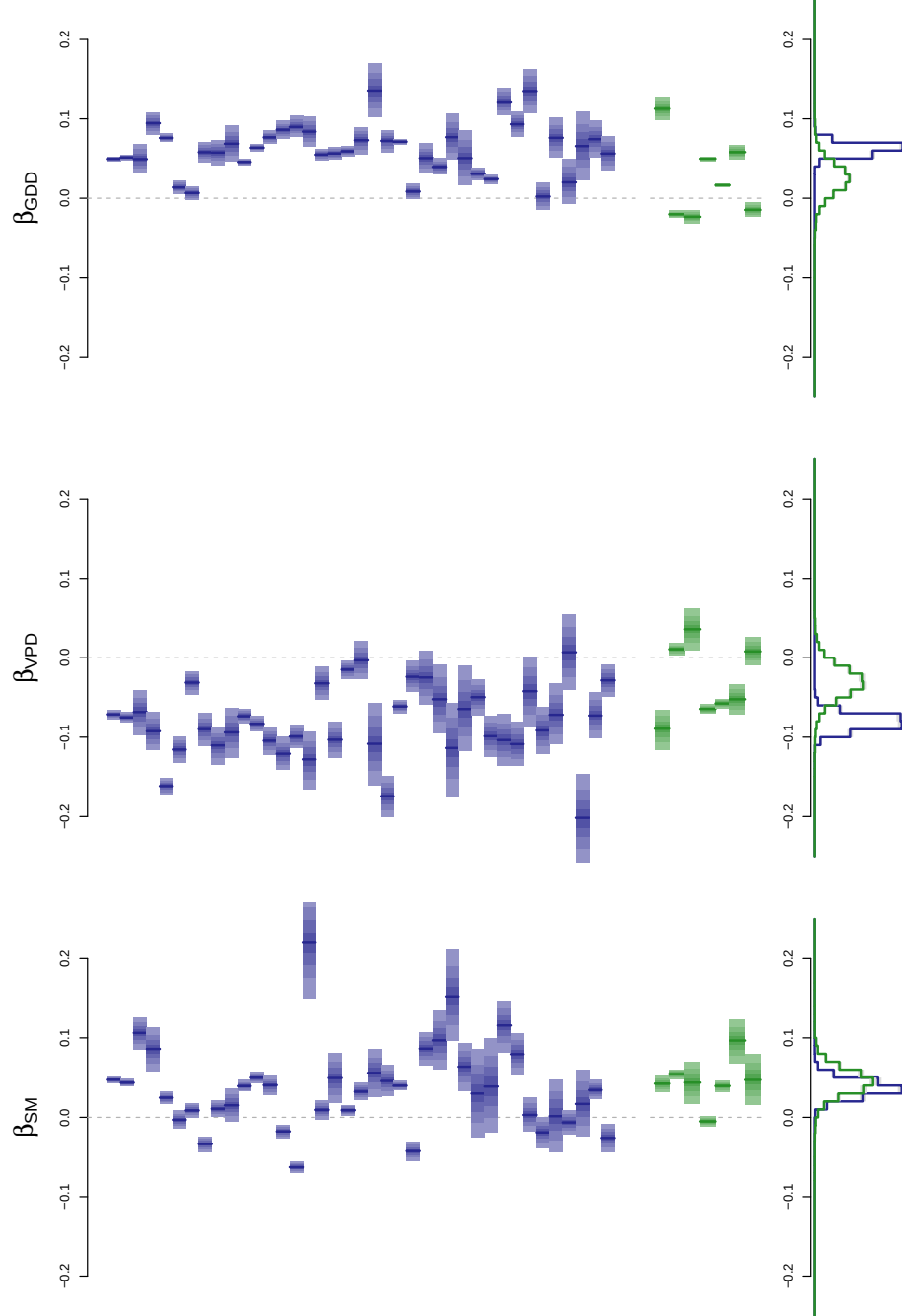
We selected three climatic predictors that are believed to have a strong impact on tree growth:

- growing degree days (GDD, between)
- soil moisture (SM,) – which is mostly driven by winter precipitation
- vapor pressure deficit (VPD,)

i Note

We extracted climate from `wldas`, which is available from 1980, at a X-m resolution. Since we plan to run our model from 1900 onward, we will also use `prism` (with simpler predictors, likely GDD and winter precipitation only).

These 3 predictors are combined within a simple linear regression. We allow the effect of each predictor to **vary across species** (i.e. we estimate species-specific slopes β). This allows to identify different sensitivities between species, and differentiate between Gymnosperms (in blue) and Angiosperms (in green).



In the left panel of this figure, each bar represents the posterior distribution of a species-specific slope. The right part shows the clade-level estimate (i.e. Gymnosperms or Angiosperms).

2.2.2 Shared residual variations (stand-level)

The above predictors are not sufficient to capture the **shared variations** of growth across trees within the same stand.

We added a short-term GP, initially with the goal to capture 3 to 5-year patterns. But the model consistently finds some yearly residual variation. In other words, the length scale of this GP is consistently around 1 year.

i Note

While the long-term GP was at the individual level (each tree gets its own individual GP, determined by a species-specific length scale), this short-term GP is shared across all trees in the same stand – and we estimate only one length scale across all stands.

2.2.3 Shocks!

Show that it's on drought years, but some other years are similarly dry.